

PROJECT SYNOPSIS

SkyCast – Flight Delay Prediction & Risk Analytics

1. Introduction

Air travel delays are a persistent and costly problem across the aviation industry, affecting airlines, airports, and passengers alike. A single delayed flight can cascade into disruptions across connecting flights, crew schedules, and gate assignments, resulting in significant operational and financial losses. This project, SkyCast, applies machine learning techniques to predict flight delays before they occur, enabling proactive decision-making rather than reactive damage control.

2. Problem Statement

Given historical flight data such as airline, route, scheduled departure time, and weather conditions, the objective is to build a system that can (a) classify whether a given flight is likely to be delayed, (b) estimate the expected duration of the delay in minutes, and (c) flag routes or time periods exhibiting abnormal delay behavior compared to historical norms.

3. Objectives

To analyze historical flight data and identify key factors contributing to delays.

To build a binary classification model that predicts the likelihood of a flight being delayed.

To build a regression model that estimates the expected delay duration for delayed flights.

To apply anomaly detection to identify routes or time periods with unusual delay patterns.

To visualize delay trends by airline, airport, route, and time of day through an interactive dashboard.

4. Scope of the Project

The project focuses on domestic flight delay prediction using historical airline operations data. It covers data cleaning, exploratory data analysis, feature engineering, model training and evaluation, anomaly detection, and the development of a dashboard for presenting results. The system is intended as a decision-support and analytical tool, not a real-time operational system for live airline scheduling.

5. Dataset

The project uses a publicly available airline delay and cancellation dataset (e.g., US DOT flight performance data available via Kaggle), containing fields such as airline, origin and destination airports, scheduled and actual departure/arrival times, delay duration, and cause of delay.

6. Methodology

Data Cleaning: Handling missing values, duplicate records, and inconsistent timestamps.

Feature Engineering: Extracting departure hour, day of week, month, and computing historical average delay by airline and route.

Classification: Training Logistic Regression, Random Forest, and XGBoost models to predict delay probability.

Regression: Estimating delay duration in minutes for flights predicted to be delayed.

Anomaly Detection: Applying Isolation Forest to identify routes or time windows with abnormal delay behavior.

Visualization: Building charts and an interactive dashboard summarizing delay trends and model outputs.

7. Technology Stack

Programming Language	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn, Plotly
Machine Learning	Scikit-learn, XGBoost
Anomaly Detection	Isolation Forest (Scikit-learn)
Dashboard	Streamlit
Version Control	Git and GitHub

8. Evaluation Metrics

Classification: Precision, Recall, F1-score, ROC-AUC score.

Regression: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 score.

Anomaly Detection: Precision and recall on flagged anomalous routes/periods.

9. Expected Outcome

The project is expected to deliver a working machine learning pipeline capable of predicting flight delay risk and duration with reasonable accuracy, along with an interactive dashboard that visualizes delay trends by airline, airport, and time, and highlights routes with abnormal delay patterns requiring attention.

10. Conclusion

SkyCast demonstrates the practical application of classification, regression, and anomaly detection techniques to a real-world operational problem faced by the aviation industry. By combining predictive modeling with clear visual analytics, the project provides actionable insights that could support proactive delay management in a real airline or airport setting.